

2.1 Mean vectors and covariance matrices

Random vectors enter regression in two different ways. The first is the response vector. In the model

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon},$$

$\mathbf{Y} = (Y_1, \dots, Y_n)\mathbf{T}$ has one entry per observational unit, and so does the unobservable error $\boldsymbol{\varepsilon}$. The assumptions that do the real work in Chapter 5, Chapter 6 and Chapter 7 are not about the shape of any distribution. They are statements about moments: $\mathbf{E}(\boldsymbol{\varepsilon}) = \mathbf{0}$, and the errors have equal variances and are uncorrelated. The second way is the measurement vector. When several quantities are recorded on the same unit, such as income, age and years of schooling, the unit contributes a vector whose components are usually correlated and measured in different units. Random regressors, errors-in-variables models and multivariate responses are all built from such vectors.

Both uses need the same toolkit: expectation and covariance as operations on vectors and matrices, their behaviour under linear maps, and the expected values of quadratic forms. Throughout, a random vector *has finite second moments* if every component has finite variance.

2.1.1 Random matrices and their expectation

Definition 2.1.1 — Expectation of a random matrix

A **random matrix** $\mathbf{Z} = (Z_{ij})$ is an $m \times k$ array of random variables. If every Z_{ij} has a finite expectation, the **expectation** of $\mathbf{Z}$ is the $m \times k$ matrix of expectations,

$$\mathbf{E}(\mathbf{Z}) = (\mathbf{E}(Z_{ij})).$$

A random vector is the case $k = 1$.

The definition works entry by entry, so every rule that is linear in the entries carries over at once.

Proposition 2.1.1 — Linearity of expectation

Let $\mathbf{Z}$ and $\mathbf{Z}_2$ be $m \times k$ random matrices with integrable entries, and let $\mathbf{A}$ ($l \times m$), $\mathbf{B}$ ($k \times s$) and $\mathbf{C}$ ($l \times s$) be constant matrices. Then

- (a) $\mathbf{E}(\mathbf{AZB} + \mathbf{C}) = \mathbf{A} \mathbf{E}(\mathbf{Z}) \mathbf{B} + \mathbf{C}$;
- (b) $\mathbf{E}(\mathbf{Z} + \mathbf{Z}_2) = \mathbf{E}(\mathbf{Z}) + \mathbf{E}(\mathbf{Z}_2)$ and $\mathbf{E}(\mathbf{ZT}) = \mathbf{E}(\mathbf{Z})\mathbf{T}$;
- (c) if $m = k$, then $\mathbf{E} \operatorname{tr}(\mathbf{Z}) = \operatorname{tr} \mathbf{E}(\mathbf{Z})$.

Proof. The (i, j) entry of $\mathbf{AZB} + \mathbf{C}$ is $\sum_r \sum_t a_{ir} Z_{rt} b_{tj} + c_{ij}$, a finite linear combination of integrable random variables with constant coefficients. Its expectation is $\sum_r \sum_t a_{ir} \mathbf{E}(Z_{rt}) b_{tj} + c_{ij}$, which is the (i, j) entry of $\mathbf{A} \mathbf{E}(\mathbf{Z}) \mathbf{B} + \mathbf{C}$. Parts (b) and (c) are the same argument applied to $Z_{ij} + Z_{2,ij}$, to Z_{ji} , and to $\sum_i Z_{ii}$. ■

Two warnings go with this proposition. The multipliers must be constant: if $\mathbf{A}$ is random, $\mathbf{E}(\mathbf{AZ})$ is generally not $\mathbf{E}(\mathbf{A})\mathbf{E}(\mathbf{Z})$. And the expectation of a product of random matrices is generally not the product of expectations. It is when every entry of the first is independent of every entry of the second, because then $\mathbf{E}(Z_{ir}W_{rj}) = \mathbf{E}(Z_{ir})\mathbf{E}(W_{rj})$ for each term.

2.1.2 Mean vector and covariance matrix

Definition 2.1.2 — Mean vector and covariance matrices

Let $\mathbf{Y}$ be a $p \times 1$ random vector with finite second moments. Its **mean vector** is $\boldsymbol{\mu} = \mathbf{E}(\mathbf{Y})$, and its **covariance matrix** is the $p \times p$ matrix

$$\text{Cov}(\mathbf{Y}) = \mathbf{E}[(\mathbf{Y} - \boldsymbol{\mu})(\mathbf{Y} - \boldsymbol{\mu})^T],$$

whose (i, j) entry is $\sigma_{ij} = \text{Cov}(Y_i, Y_j)$ and whose diagonal holds the variances $\sigma_{ii} = \text{Var}(Y_i)$. If $\mathbf{U}$ ($p \times 1$) and $\mathbf{V}$ ($q \times 1$) both have finite second moments, their **cross-covariance matrix** is the $p \times q$ matrix

$$\text{Cov}(\mathbf{U}, \mathbf{V}) = \mathbf{E}[(\mathbf{U} - \mathbf{E}\mathbf{U})(\mathbf{V} - \mathbf{E}\mathbf{V})^T],$$

with (i, j) entry $\text{Cov}(U_i, V_j)$.

The expectations exist: $|U_i V_j| \leq \frac{1}{2}(U_i^2 + V_j^2)$, so finite second moments make every product integrable. We often write $\boldsymbol{\Sigma}$ for $\text{Cov}(\mathbf{Y})$, and $\text{Cov}(\mathbf{Y}) = \text{Cov}(\mathbf{Y}, \mathbf{Y})$. Some books write $\text{Var}(\mathbf{Y})$ or $D(\mathbf{Y})$ for the covariance matrix. We keep Var for scalars. The symbol Cov then has three uses: $\text{Cov}(Y_i, Y_j)$ is a number, $\text{Cov}(\mathbf{Y})$ is a symmetric square matrix, and $\text{Cov}(\mathbf{U}, \mathbf{V})$ is a rectangular matrix that is usually not symmetric even when $p = q$.

Proposition 2.1.2 — Elementary properties

Let $\mathbf{Y}, \mathbf{U}, \mathbf{V}$ have finite second moments, with $\boldsymbol{\mu} = \mathbf{E}(\mathbf{Y})$ and $\boldsymbol{\Sigma} = \text{Cov}(\mathbf{Y})$, and let $\mathbf{a}, \mathbf{b}$ be constant vectors of matching sizes. Then:

- (a) $\boldsymbol{\Sigma}$ is symmetric;
- (b) $\text{Cov}(\mathbf{U}, \mathbf{V}) = \mathbf{E}(\mathbf{U}\mathbf{V}^T) - \mathbf{E}(\mathbf{U})\mathbf{E}(\mathbf{V})^T$, and in particular $\boldsymbol{\Sigma} = \mathbf{E}(\mathbf{Y}\mathbf{Y}^T) - \boldsymbol{\mu}\boldsymbol{\mu}^T$;
- (c) $\text{Cov}(\mathbf{V}, \mathbf{U}) = \text{Cov}(\mathbf{U}, \mathbf{V})^T$;
- (d) $\text{Cov}(\mathbf{U} + \mathbf{a}, \mathbf{V} + \mathbf{b}) = \text{Cov}(\mathbf{U}, \mathbf{V})$;
- (e) $\mathbf{E}[(\mathbf{Y} - \mathbf{a})(\mathbf{Y} - \mathbf{a})^T] = \boldsymbol{\Sigma} + (\boldsymbol{\mu} - \mathbf{a})(\boldsymbol{\mu} - \mathbf{a})^T$, and hence $\mathbf{E}\|\mathbf{Y} - \mathbf{a}\|^2 = \text{tr}(\boldsymbol{\Sigma}) + \|\boldsymbol{\mu} - \mathbf{a}\|^2$.

Proof.

- (a) and (c) hold because $(\mathbf{uv}^T)^T = \mathbf{vu}^T$ and expectation commutes with transposition. For (b), expand $(\mathbf{U} - \boldsymbol{\mu}_U)(\mathbf{V} - \boldsymbol{\mu}_V)^T = \mathbf{U}\mathbf{V}^T - \mathbf{U}\boldsymbol{\mu}_V^T - \boldsymbol{\mu}_U\mathbf{V}^T + \boldsymbol{\mu}_U\boldsymbol{\mu}_V^T$ and take expectations with Proposition 2.1.1. The two middle terms each give $-\boldsymbol{\mu}_U\boldsymbol{\mu}_V^T$. (d) holds because adding a constant does not change the centred vectors. For (e), write $\mathbf{Y} - \mathbf{a} = (\mathbf{Y} - \boldsymbol{\mu}) + (\boldsymbol{\mu} - \mathbf{a})$ and expand. The cross terms contain the factor $\mathbf{E}(\mathbf{Y} - \boldsymbol{\mu}) = \mathbf{0}$ and vanish. The trace identity follows from

$\|\mathbf{v}\|^2 = \text{tr}(\mathbf{v}\mathbf{v}^T)$ and Proposition 2.1.1(c). ■

Part (e) is the vector form of “mean squared error equals variance plus squared bias”. It shows that the mean is the constant vector closest to $\mathbf{Y}$ in expected squared distance, and that the minimal value is $\text{tr}(\mathbf{\Sigma}) = \sum_i \text{Var}(Y_i)$. The matrix statement is stronger than the trace statement. It says that the second-moment matrix about any point exceeds $\mathbf{\Sigma}$ by a nonnegative definite matrix of rank at most one.

2.1.3 Uncorrelated is not independent

Random vectors $\mathbf{U}$ and $\mathbf{V}$ are **uncorrelated** if $\text{Cov}(\mathbf{U}, \mathbf{V}) = \mathbf{0}$. If $\mathbf{U}$ and $\mathbf{V}$ are independent, then $\mathbf{E}(U_i V_j) = \mathbf{E}(U_i)\mathbf{E}(V_j)$ for all i, j , so they are uncorrelated. The converse is false, and it fails in the most innocent-looking cases.

Example. (A point on a circle)

Let Θ be uniform on $[0, 2\pi)$ and $\mathbf{Y} = (\cos \Theta, \sin \Theta)^T$. Then $\mathbf{E}(\cos \Theta) = \mathbf{E}(\sin \Theta) = 0$, $\mathbf{E}(\cos^2 \Theta) = \mathbf{E}(\sin^2 \Theta) = \frac{1}{2}$, and $\mathbf{E}(\cos \Theta \sin \Theta) = \frac{1}{2}\mathbf{E}(\sin 2\Theta) = 0$. So $\boldsymbol{\mu} = \mathbf{0}$ and $\text{Cov}(\mathbf{Y}) = \frac{1}{2}\mathbf{I}_2$, exactly the moments of two independent coordinates. Yet $Y_1^2 + Y_2^2 = 1$: once Y_1 is known, Y_2 is known up to sign. Covariance measures *linear* association only, and the dependence here is entirely nonlinear.

For jointly normal vectors, zero covariance and independence coincide (Theorem 3.4.1). That equivalence rests on joint normality, not on normal marginals, and it is one of the main reasons normal theory is so tractable.

Scalar summaries. The **total variance** $\text{tr}(\mathbf{\Sigma})$ ignores correlation; the **generalized variance** $\det(\mathbf{\Sigma})$ does not. If $\lambda_1, \dots, \lambda_p$ are the eigenvalues of $\mathbf{\Sigma}$ (Theorem 1.7.2), these are $\sum_i \lambda_i$ and $\prod_i \lambda_i$, and both are unchanged by a rotation $\mathbf{Y} \mapsto \mathbf{Q}\mathbf{Y}$. The determinant can be tiny even when every variance is large, if one eigenvalue is near zero. Theorem 2.2.3 shows that a zero eigenvalue means some linear combination of $\mathbf{Y}$ does not vary at all.

2.1.4 Exercises

2.1.4.1 A. Check your understanding

Exercise A1.

Let $\mathbf{Z}$ be an $n \times p$ random matrix whose entries are independent, with mean 0 and variance σ^2 . Show that $\mathbf{E}(\mathbf{Z}^T \mathbf{Z}) = n\sigma^2 \mathbf{I}_p$ and $\mathbf{E}(\mathbf{Z} \mathbf{Z}^T) = p\sigma^2 \mathbf{I}_n$. For a constant $p \times p$ matrix $\mathbf{B}$, find $\mathbf{E}(\mathbf{Z} \mathbf{B} \mathbf{Z}^T)$.

2.1.4.2 B. Practice

Exercise B1.

Let X have a distribution symmetric about 0 with $\text{Var}(X) = 1$ and $\mathbf{E}X^4 < \infty$, let S be independent of X with $\Pr(S = 1) = \Pr(S = -1) = \frac{1}{2}$, and put $\mathbf{Y} = (X, SX)\mathbf{T}$. Show that the two components have the same distribution and are uncorrelated, and that $\text{Cov}(Y_1^2, Y_2^2) = \text{Var}(X^2)$. Deduce that Y_1 and Y_2 are independent iff X is itself a random sign. (Chapter 3 takes X standard normal.)

Solution. Since S is independent of X and X is symmetric, $\Pr(SX \leq y) = \frac{1}{2} \Pr(X \leq y) + \frac{1}{2} \Pr(-X \leq y) = \Pr(X \leq y)$. By independence, $\text{Cov}(X, SX) = \mathbf{E}(S)\mathbf{E}(X^2) = 0$. Both squares equal X^2 , so $\text{Cov}(Y_1^2, Y_2^2) = \text{Var}(X^2)$. Functions of independent variables are independent, hence uncorrelated, so $\text{Var}(X^2) > 0$ makes Y_1 and Y_2 dependent. If $\text{Var}(X^2) = 0$, then $X^2 = \mathbf{E}X^2 = 1$ with probability one, and by symmetry $X = \pm 1$ with probability $\frac{1}{2}$ each. Then $\Pr(Y_1 = a, Y_2 = b) = \Pr(X = a) \Pr(S = ab) = \frac{1}{4}$ for all signs a, b , which is the product of the marginal probabilities, as in Example (Pairwise but not mutually independent).

Exercise B2.

Show that $\mathbf{E}[(\mathbf{Y} - \mathbf{a})(\mathbf{Y} - \mathbf{a})\mathbf{T}] - \text{Cov}(\mathbf{Y})$ is nonnegative definite for every constant $\mathbf{a}$, with equality iff $\mathbf{a} = \boldsymbol{\mu}$. Deduce that $\mathbf{a} = \boldsymbol{\mu}$ minimizes $\mathbf{E}[(\mathbf{Y} - \mathbf{a})\mathbf{T}\mathbf{N}(\mathbf{Y} - \mathbf{a})]$ for every positive definite $\mathbf{N}$.